

Course Title: CLIENT SERVER AND WEB DEVELOPMENT

Course code: CEF 344

Exercise 1: Student Result Processing System (70 Marks)

Develop a web application that computes and displays student academic results.

Functional Requirements

Data Entry: Enter Student Name, Course Code, Continuous Assessment Score (/30), Examination Score (/70) (20 Marks)

Grade Computation: Calculate Total Score, Grade, Remark using the table below. (20 marks)

Score	Grade
70 - 100	A
60 - 69	B
50 - 59	C
45 - 49	D
0 - 44	F

Display results : Display all processed results on a table. (15marks)

Search Functionality : Search students by name. (5 Marks)

Validation : Prevent Empty fields, Scores above 100, Negative values. (10 Marks)

Exercise 2: Student Registration and Management System (70 Marks)

Develop a web application that allows a department secretary to register and manage students.

Functional Requirements

Student Registration : The system should allow users to Enter Student matriculation number, Enter Full Name, Enter Email Address, Select Gender, Select Department. (20 Marks)

Display Registered Students : Display all registered students in a table, Each row should contain all student information. (15 Marks)

Search Functionality : Search students by name, Matching records should be displayed dynamically. (10 Marks)

Edit and Delete Operations : Modify student information, delete a student record. (15 Marks)

Form Validation : No field should be left empty, email should be validated, student ID should be unique.(10 Marks)

Exercise 3: Library Book Management System (70 Marks)

Develop a web application that allows a librarian to manage books.

Functional Requirements

Book Registration : Capture Book ID, Title, Author, Category **(20 Marks)**

Display Books : Display all books in a table. **(15 Marks)**

Search Books :Search by title. **(10 Marks)**

Edit and Delete Books : Allow updating and removing books. **(15 Marks)**

Validation: Ensure all fields are completed and duplicate Book IDs are not accepted. **(10 Marks)**

Exercise 4: Library Book Management System (70 Marks)

Develop a web application that allows a librarian to manage books.

Functional Requirements

Book Registration

Capture Book ID, Title, Author, Category **(20 Marks)**

Display Books : Display all books in a table. **(15 Marks)**

Search Books : Search by title. **(10 Marks)**

Edit and Delete Books : Allow updating and removing books. **(15 Marks)**

Validation : Ensure all fields are completed and duplicate Book IDs are not accepted. **(10 Marks)**

Exercise 5: Restaurant Menu Ordering System

Create a simple online restaurant ordering application.

Functional Requirements

Menu Display : Display at least food Name, price, image **(15 Marks)**

Add-to-Order Feature : Users can add items to an order list. **(15 Marks)**

Bill : Display selected Items, quantity, total cost **(15 Marks)**

AJAX Menu Loading : Load menu items from a JSON file using AJAX. **(15 Marks)**

User Interface : Responsive and user-friendly design. **(10 Marks)**

Note: Aside from the functional requirements, you are expected to present your web applications with good UIs, suitable usability, and incorporate all the elements discussed in class. Students should work in their groups on the exercises as distributed.

Groups	Exercise
1, 2,15	1
3, 13, 14	2
4, 5, 12	3
6, 10, 11	4
7, 8, 9	5